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MACHINE LEARNING POWERED MOSQUITO POPULATION CONTROL SYSTEM FOR SEWAGE NETWORKS





BACKGROUND



HEALTH CRISIS

263 million cases of mosquito-borne diseases

01



UNCONVENTIONAL MOSQUITO PREVENTION METHODS

02



SUSTAINABILITY AND SELF-OPERATING SYSTEM

03

PROJECT OBJECTIVE



Develop

an autonomous, ML-powered system for mosquito control in urban sewage networks.



Target

breeding and resting sites to break the mosquito life cycle



Provide

a scalable, chemical-free, and data-driven alternative to fogging.

PROJECT COMPONENTS

Component	Rationale for Selection
Raspberry Pi 5	To develop the Machine-Learning function using YOLOv5 computer vision model
Raspberry Pi Camera	To identify mosquitos inside sewage networks
Water Sensor	To prevent electrical components from interacting with increasing water level, thereby increasing its safety and lifespan
Relay	To connect Raspberry Pi with the mosquito racket while maintaining its lifespan and reliability
Used Mosquito Racket	To omit mosquitos detected by the Raspberry Pi Camera

MACHINE LEARNING MODEL

Algorithm Architecture and Rationale

Accuracy

The system uses a lightweight CNN-based YOLO model (e.g., YOLOv5) for fast and accurate mosquito detection from thermal camera feeds.

01

Precision

YOLO’s balance of speed and precision makes it ideal for real-time detection on edge devices.

02

Effectiveness

Previous studies report 95–98% accuracy using YOLOv5 and other CNN models (Faster R-CNN, VGG16), proving the effectiveness of deep learning for mosquito identification.

03

Model Training and Edge Optimization

01

Process

Model development follows a two-stage process: training and optimization, using a custom dataset of images captured and labeled.

02

Feature

The CNN learns distinct features of mosquitoes—such as the shape, and size—through annotated training data.

03

Challenge

The main challenge is distinguishing mosquitoes from other objects, requiring a carefully curated and diverse dataset to ensure accuracy.

CNN = Convolutional Neural Network

PERFORMANCE METRICS

Metric	Value	Description
mAP50-95(B)	~0.85	Mean Average Precision, the primary metric for object detection accuracy across various confidence thresholds.
Precision	~0.92	Of all detections made, the percentage that were actually mosquitoes (low false positives).
Recall	~0.92	Of all actual mosquitoes present, the percentage that were correctly detected (low false negatives).
Inference Time	~150 ms	The time taken for the model to process one frame.
Frames Per Second (FPS)	~6-7 FPS	The real-time processing speed (1000ms / 150ms), sufficient for detecting resting or slow-moving insects.

INNOVATION & ADVANTAGES

Autonomy

Operates automatically without dependance on human work



01

Targeted Eradication

Directly targeted at mosquitos and its breeds ensuring sustainability for its surrounding and ecosystem



02

Chemical-Free Operation

Avoids environmental problems caused by chemical pollution



03

POTENTIAL FOR PUBLIC HEALTH IMPACT

01

Vector Control

The system reduces mosquito populations in city sewage areas.

02

Disease Prevention

It helps prevent dengue and malaria by cutting disease transmission.

03

Data Collection

Each detection gives time and location data for tracking.

04

Networked System

Multiple devices create a city-wide mosquito monitoring network.

05

Early Warnings

Enables early detection of mosquito hotspots before outbreaks.

06

Smart Public Health

Supports a data-driven and proactive public health system.

CONCLUSION

01

PURPOSE

Developing an autonomous, sustainable, environment-friendly eradication system

02

INNOVATION

Integrating Machine-Learning and autonomous operating system in high-voltage low-current eradication

03

KEY FINDINGS

High detection accuracy, advantage in accesibility and targeted system to omit reliance on chemicals

04

IMPACT

Establishes a foundation for next generation system from reactive to proactive and data driven system

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THANK YOU